

Fardin Ahmed

US Citizen | fahmed71@gatech.edu | 720-569-9469 | LinkedIn | fineredium1.github.io

EDUCATION

Georgia Institute of Technology
B.S. in Computer Science.

Atlanta, GA
August 2024 - May 2027

SKILLS

Languages: Python, C/C++, JavaScript, Go, Java, SQL, MATLAB, R, HTML/CSS
Frameworks: React, Node.js/Express, Django, Flask
Libraries: PyTorch, NumPy, scikit-learn, Google Test, Matplotlib, SFML, Pygame, Libgdx
Developer Tools: Git, AWS, Shell Scripting, Jira, VirtualBox, Makefile, Eclipse, Cursor
Operating Systems: Linux (Debian, Ubuntu, Red Hat), MacOS, Windows

EXPERIENCE

Undergraduate Researcher, VIP Program January 2026 – Present
Georgia Institute of Technology Atlanta, GA

- Co-authoring a research paper on a benchmark and training environment for LLM forecasting agents that separates retrieval from reasoning and evaluates probabilistic forecasts against prediction-market baselines.
- Pretrained TinyTimeMixer (TTM) by doubling hidden dimensions from 64 to 128, reducing FRED-MD forecasting MSE by 83%.
- Benchmarked Chronos Bolt Tiny, PatchTST, DeepAR, and TTM on 120+ FRED-MD macroeconomic series using RMSE, MAE, and MASE metrics via AutoGluon.

Software Engineering Intern May 2026 – August 2026
Cisco Systems, Inc. San Jose, CA

- Designed and shipped an end-to-end MCP-based CLI agent in stdlib-only Python, exposing live switch diagnostics as structured tools in air-gapped environments and running a quantized Gemma 4 2B model via llama.cpp with KV-cache quantization and memory tuning; adopted by 200+ engineers as an internal network operations tool to reduce manual CLI troubleshooting.
- Engineered a ReAct reasoning loop with self-evaluation, checkpoint backtracking, and BM25 retrieval across 1,500+ SONiC and Cisco-specific command references; distilled tool-use trajectories from a larger teacher model for LoRA fine-tuning, improving tool-calling and argument-generation accuracy from 60% to 92%.

Software Engineering Intern May 2025 – August 2025
Science Applications International Corporation (SAIC) Colorado Springs, CO

- Modernized object-oriented C++ backend modules by transitioning legacy communication protocols to TCP/IP socket-based systems; implemented 50+ Google Test unit tests, raising test coverage from 15% to 35%.
- Contributed to development and testing of critical defense software systems, including radar system components, in a high-stakes Agile environment while holding an Interim Secret clearance.

PROJECTS

Safelight April 2026 – Present
React, Node/Express, PostgreSQL

- Built end-to-end encryption for a full-stack messaging app using a zero-dependency Web Crypto module securing every message with per-message X25519 ECDH (256-bit keys), HKDF-SHA256, and AES-256-GCM in a multi-recipient sealed-envelope scheme.
- Engineered a client-side password-derived key vault using PBKDF2 (300,000+ iterations) and AES-256-GCM; added SHA-512 (5,200-round) safety-number verification and a 10-case automated suite covering tamper, wrong-key, and replay rejection.

OmniDetect January 2026 – May 2026
Python, scikit-learn, PyTorch, librosa

- Built the text-detection component of a 5-person team project that flags whether text, images, or audio was AI-generated, engineering a TF-IDF feature pipeline (single + two-word phrases, 70,000-word vocabulary) and training a LinearSVC classifier that reached 96.93% accuracy at a 0.8% false-positive rate.
- Validated generalization with leakage-safe 70/15/15 splits and 4 controlled baseline experiments (Logistic Regression vs. LinearSVC, with/without TF-IDF), sustaining 95.6% accuracy on a separate 450K+ sample unseen dataset.

ACTIVITIES & LEADERSHIP

GT Trading Club August 2026 - Present
Member

- Participate in weekly quantitative finance workshops and market discussions focused on trading strategies, financial modeling, and industry interview preparation.

GT Web Development Club August 2024 - Present
Member

- Contributed to team-based projects building campus-related web solutions using HTML, CSS, JavaScript, Node.js, and React.